Southern New Hampshire University

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Data Analysis Process Job Aid

Who should use this job aid?

This job aid will be given to the new hires of my organization. This aid will make specific references to the Storm and Crime Data Report and will enumerate the steps used prior to analysis. The new hires will be able to use my process as a model for their future work.

The goal of this job aid is to promote efficiency and improve the quality of work for new hires. If a standard example is provided for training, the quality of employees during onboarding will improve and results will be more standard.

Introduction

In this project I am a senior data analyst at a data consulting firm. I have been tasked with helping to train a group of newly hired data analysts on the data analysis processes that my organization uses to work with clients. In order to do this, I will use my current project with the Miami police department.

The police department has requested the creation of a Storm and Crime Data Report using historical data from the city of Miami for the year 2024 because the police force has made a connection between crimes and storms. This report will provide the department with information they can use to anticipate possible time frames for crimes in the future.

Section 1: Type of analysis

The police department wants to be able to be able to predict crime by analyzing the linkage between crime and weather. I was given two initial data sets from a crime table and a storm table. Both tables have multiple variables but I only need to pick out the ones I need to create the merged table for the Storm and Crime Data Report.

The date variable is important in this project and links the storm event numbers to the crime incident numbers. It is formatted differently between the two tables, so this has to be adjusted. This way when the tables merge the dates can enter under the same column. The total storm events numbers and the total crime incident numbers also need to be found.

The final visualization data analysis will allow the viewer to best come up with a conclusion whether there is link between storm incidents and crime. This visualization will allow the user to easily compare the two variables and see if any trends occur in the year 2024.

Section 2: Define Parameters and collect data

The relevant weather data variables required for analysis will include Begin_YearMonth, Begin_Day, Event_Type, state, and neighborhood. The relevant criminal data variables required for analysis will include CFSDate, Neighborhood, and Incident Type. The query that needs to be created will isolate the storm events that occurred in Florida as the weather database is larger and includes information from other states. With this information I can create the second query to pull the information of the total number of storm events throughout the different Florida neighborhoods in 2024.

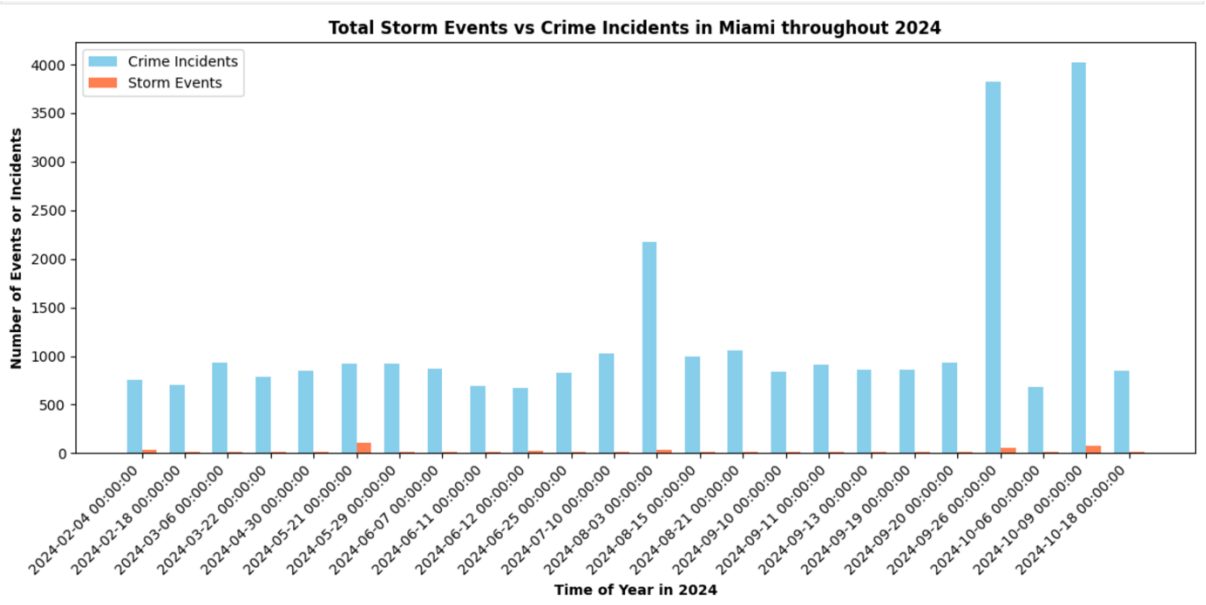
The third query will pull the total number of incident reports in Florida in 2024. This data will me to create a graph later on based on the date to compare total storm event and crime incident numbers. This way it will be easier to see if there is a correlation between the two.

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Section 3: Tool Selection

At the start of this project, I used MySQL Bench and created the sql queries mentioned above. These queries gave me the raw data I needed to move to Jupyter Notebook and use Python to perform further data analysis. The initial queries I made in Jupyter were to connect and import the information from the sql application.

I cleaned the datasets from the crime and storm databases by checking for missing values. After the cleaning step was done I merged the data tables by combining the date column and making sure the formatting was uniform. This new chart is the SCDR that the police department wanted. This report shows the date, the neighborhood in Florida, the total number of storm events on that day, and the total number of criminal incidents on the matching date. I chose to use a bar chart as the most suitable visualization to demonstrate the outcome of the data analysis. An image of this visualization can be seen below.



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Section 4: Validation

In order to validate the data, I used the query below. This query accesses the merged data table. As you see the query does validate that I am pulling information from both data sets to answer the police departments question of the number of storm events vs crime incidents in Florida in 2024.

```
]: crimes_df.rename(columns={'CrimeDay': 'StormDay'}, inplace=True)

merged_df = pd.merge(storms_df, crimes_df, how='inner', on="StormDay")
print(merged_df.head())
```

```
     StormDay NEIGHBORHOOD_x  StormCount NEIGHBORHOOD_y  Incident
0  2024-02-04     MIAMI-DADE           3        Flagami        71
1  2024-02-04     MIAMI-DADE           3   Little Haiti        47
2  2024-02-04     MIAMI-DADE           3     Model City        73
3  2024-02-04     MIAMI-DADE           3     Allapattah        83
4  2024-02-04     MIAMI-DADE           3  Coconut Grove        35
```

I also set up a query to determine the acceptable range of parameters for the separate data tables as can be seen below. This way the numbers on the chart can confirmed as being accurate and within range.

```
#stats for number of storm events and crime incidents in Florida
stats_storm = storms_df['StormCount'].describe()
print(f'STATS for storm events: \n{stats_storm}')

print()

stats_crime = crimes_df['Incident'].describe()
print(f'STATS for crime incidents: \n{stats_crime}')
```

```
STATS for storm events:
count    33.000000
mean      1.333333
std       1.266557
min       1.000000
25%       1.000000
50%       1.000000
75%       1.000000
max       8.000000
Name: StormCount, dtype: float64

STATS for crime incidents:
count    4758.000000
mean       66.125473
std        28.312738
min        14.000000
25%        44.000000
50%        60.000000
75%        85.000000
max       176.000000
Name: Incident, dtype: float64
```